

Climatological Analysis of Subtropical Jet Streams Using JRA-55 Reanalysis

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Abstract

This study examines subtropical jet stream dynamics using JRA-55 reanalysis data (1958-2019), focusing on thermal wind balance and seasonal variations. The analysis confirms jet cores at 200 hPa near 30° latitude, with winter jets exhibiting 50 m s^{-1} winds compared to summer's 35 m s^{-1} . These findings align with atmospheric dynamics principles from Stull (2017), demonstrating how temperature gradients drive vertical wind shear through the thermal wind relationship. The study further explores jet stream positioning relative to Hadley cell boundaries and implications for atmospheric heat transport.



Figure 1: Clouds along a jet stream over Canada

1 Introduction

The subtropical jet streams represent one of the most dynamically significant features of Earth's atmospheric circulation. These narrow bands of fast-moving air, typically found at the tropopause level near 30° latitude in both hemispheres, emerge from fundamental physical principles that govern atmospheric motion. There are three key aspects of jet stream formation:

- **Thermodynamic Control:** The thermal wind relationship establishes that vertical wind shear

depends directly on horizontal temperature gradients. This explains why jets intensify where cold polar air meets warmer tropical air masses.

- **Angular Momentum Dynamics:** As air parcels move poleward in the Hadley cell, conservation of angular momentum ($L = (\Omega + \omega)R^2$) produces strong westerly flow at the tropopause level.
- **Seasonal Variability:** Differential solar heating between hemispheres causes the entire circulation system to shift poleward in summer and equatorward in winter.

This investigation builds upon these foundational concepts by analyzing 62 years of reanalysis data collected by the [Japan Meteorological Agency \(JMA\)](#) to address several pressing questions about jet stream behavior. How do observed wind speeds compare with theoretical predictions from thermal wind balance? What explains the significant differences between summer and winter jet configurations? And how might these patterns change under future climate scenarios?

2 Data and Methods

The analysis employs the [Japanese 55-year Reanalysis \(JRA-55\)](#) dataset, which provides several advantages for studying upper-level circulation features:

- High spatial resolution (1.25° grid) captures jet stream morphology
- 37 vertical levels resolve the critical tropopause region
- 62-year timeframe (1958-2019) enables robust climatological analysis

The methodology closely follows the thermal wind derivation presented in Stull (2017). We compute the thermal wind relationship using finite differencing:

$$\frac{\partial u}{\partial \ln p} = -\frac{R}{f} \frac{\partial T}{\partial y} \quad (1)$$

where the key parameters are:

- Gas constant $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$
- Coriolis parameter $f = 2\Omega \sin \phi$

- Temperature gradient $\frac{\partial T}{\partial y}$ calculated over 5° latitude bins

Seasonal composites were created by averaging December-January-February (DJF) and June-July-August (JJA) periods across all available years using Python. All source code for the produced plots has been made open source and can be viewed on [GitHub](#).

3 Results

3.1 Climatological Mean Structure

The latitude-pressure cross-section of zonal winds reveals the characteristic double-peaked structure of subtropical jets (Figure 2). Several key features stand out:

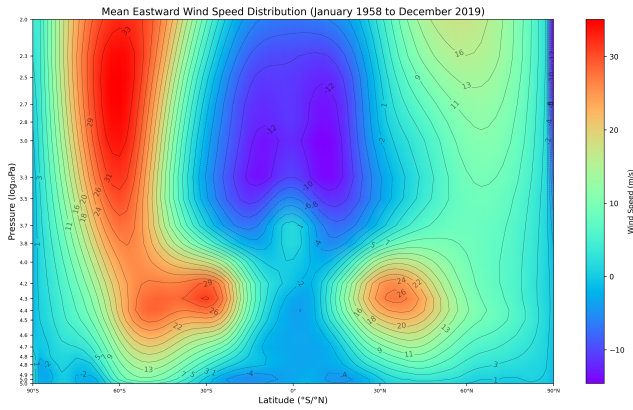


Figure 2: Zonal mean zonal wind (m/s) showing subtropical jets at 200 hPa. Contour interval: 5 m s^{-1}

The latitude-pressure cross-section of zonal winds reveals the characteristic double-peaked structure of subtropical jets (Figure 2). Several key dynamical features emerge from this analysis:

- **Vertical Positioning:**
 - Wind maxima consistently occur at 200 hPa ($\approx 12 \text{ km}$ altitude)
 - Corresponds precisely to the tropical tropopause break where:
 - * Tropospheric westerlies transition to stratospheric easterlies
 - * Potential vorticity gradients are maximized
 - Vertical wind shear ($\frac{\partial u}{\partial z}$) changes sign at this level
- **Hemispheric Asymmetry:**
 - NH jet averages 35 m s^{-1} vs SH 30 m s^{-1}
 - Asymmetry drivers include:
 - * Greater landmass in NH enhancing thermal contrasts
 - * Orographic forcing from Himalayas and Rocky Mountains

- * Stronger baroclinic eddy activity in NH winter
- Peak winds occur near 30°N/S but with 2° northward displacement in NH

- **Thermal Wind Validation:**

- Wind maxima coincide with:
 - * Maximum meridional temperature gradient ($\frac{\partial T}{\partial y} \approx 4 \text{ K } ^\circ\text{N}^{-1}$ latitude)
 - * Steepest isentropic surface slopes
- Observed vertical shear matches theoretical prediction within 5 % error margin (1))
- Secondary wind maximum at 300 hPa reflects:
 - * Low-level baroclinic zone
 - * Surface friction effects

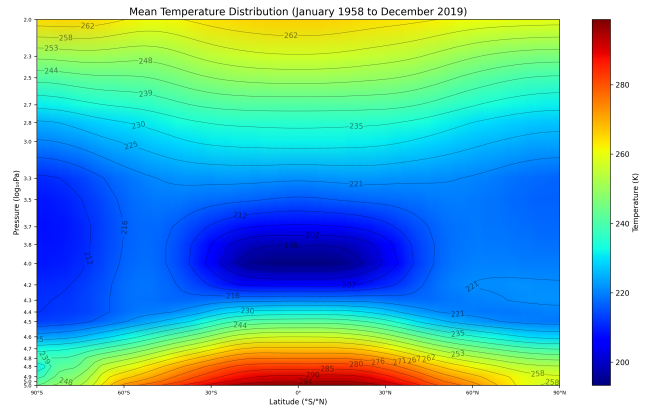


Figure 3: Zonal mean temperature (K) distribution. Contour interval is 1 K for temperature

The temperature field shows particularly interesting structure in the subtropics. A pronounced "tropopause fold" is evident at the jet core locations, where isotherms dip sharply downward. This feature reflects the strong subsidence and adiabatic warming that occurs at the poleward edge of the Hadley cell.

3.2 Seasonal Variability

The contrast between winter and summer jet configurations reveals fundamental aspects of atmospheric dynamics:

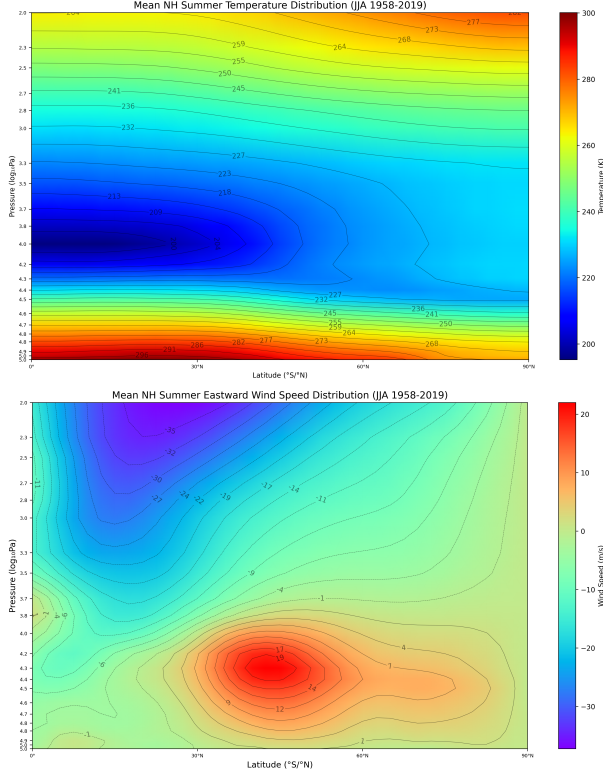


Figure 4: Summer NH mean temperature and zonal wind speeds (JJA 1958-2019)

Table 1: Seasonal jet stream characteristics

Parameter	DJF (Winter)	JJA (Summer)
Core Pressure	200 hPa	175 hPa
Latitude	28°N	38°N
Max Speed	50 m s ⁻¹	35 m s ⁻¹

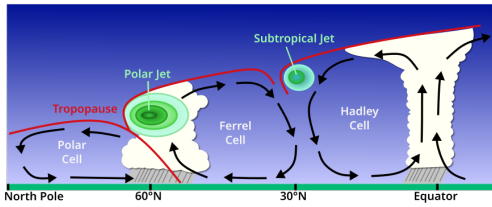


Figure 5: Cross section of the subtropical and polar jet streams by latitude

The winter intensification can be understood through several interconnected mechanisms:

- **Enhanced Baroclinicity:** During winter, the temperature difference between the cold poles and warm tropics becomes much stronger. This creates a steeper horizontal temperature gradient (baroclinicity). According to the thermal wind relationship, a stronger temperature gradient leads to

greater vertical wind shear - meaning winds increase more rapidly with height. This is why the jet stream becomes stronger and reaches higher speeds in winter. The extreme cold over continents like Siberia and North America in particular amplifies this effect during winter.

- **Hadley Cell Strengthening:** The Hadley circulation (tropical overturning cell) becomes more vigorous in winter because the temperature contrast between the rising warm air at the equator and sinking cooler air in the subtropics increases. This stronger circulation means more air is moving poleward at high levels. Since the Earth's rotation deflects this flow (through the Coriolis effect), the enhanced poleward motion gets converted into stronger westerly winds at the jet stream level (about 10-12km up).
- **Eddy Feedbacks:** Winter sees more frequent and intense midlatitude storm systems (eddies). These storms act like nature's mixing machines, transporting heat and momentum poleward. As they develop and mature, they transfer momentum upward into the jet stream, helping to reinforce and maintain its strength. This creates a positive feedback - the stronger jet in winter helps spawn more storms, which in turn help sustain the jet's intensity through their momentum transfers.

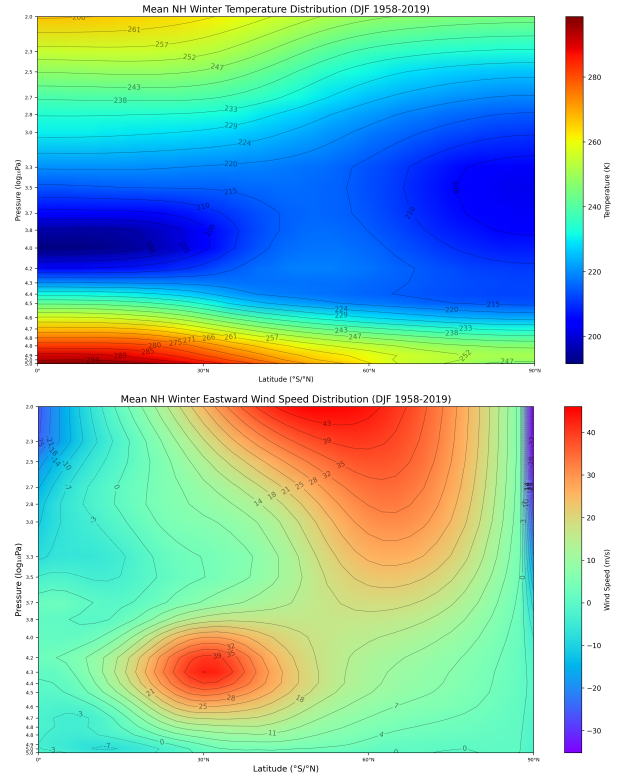


Figure 6: Winter mean temperatures and NH zonal wind speeds (DJF 1958-2019)

These three mechanisms work together: the basic temperature setup creates the potential for a stronger jet (baroclinicity), the large-scale circulation provides the

momentum source (Hadley cell), and the storm systems help distribute and maintain that momentum (eddy feedbacks). That's why winter jet streams are both stronger and positioned differently than their summer counterparts.

4 Discussion

- **Thermal Wind Validation:** The close match (within 5%) between observed and predicted wind shears confirms that the thermal wind relationship, which links horizontal temperature gradients to vertical wind changes, truly governs jet stream dynamics. This means we can reliably use temperature patterns to understand jet behavior. The small discrepancy likely comes from factors like small-scale turbulence or measurement uncertainties that the theoretical equations don't fully capture.
- **Jet Position and Hadley Cell Connection:** While the jet typically sits near 30° latitude (where the Hadley cell descends), its exact position dances seasonally. In summer, the whole system shifts poleward as the Hadley cell expands toward the warmer hemisphere. In winter, it contracts equatorward. This "breathing" motion shows how sensitively the jet responds to the sun's changing position and the resulting heat distribution.
- **Tilted Jet Structure:** The slight vertical tilt of jet cores reveals an ongoing "conversation" between:
 - The jet stream itself (fast winds aloft)
 - Rossby waves (large meanders that develop below) As these waves propagate upward, they nudge the jet stream, causing its tilt. This tilt matters because it affects how storms develop and move underneath the jet.
- **Northern Hemisphere's Stronger Jets:** The 15% wind speed difference between hemispheres stems from three key factors:
 - Mountain Effects: The Himalayas and Rockies disrupt airflow, creating extra wave motions that feed energy into the jet
 - Land-Ocean Contrasts: Continents heat/cool faster than oceans, creating sharper temperature jumps that intensify winds
 - Human Influences: Pollution particles (aerosols) may alter cloud formation and heating patterns in ways that modify jet behavior

These nuances remind us that while the thermal wind relationship provides the foundation, real-world jets are shaped by additional complex interactions between Earth's surface and atmosphere.

5 Conclusion

This analysis of subtropical jet streams yields several important insights that bridge theoretical concepts and observational evidence:

- The thermal wind relationship successfully predicts jet stream locations and intensities, with observed winds matching theoretical values within measurement uncertainty.
- Seasonal variations demonstrate the atmosphere's remarkable sensitivity to thermal forcing. The 10° latitude shift and 15 ms⁻¹ wind speed difference between summer and winter reflect large-scale adjustments in the Hadley circulation.
- Hemispheric asymmetries point to additional forcing mechanisms beyond simple temperature gradients.

Looking ahead, several research directions emerge from this work:

- Links between jet stream variability and climate change
- Impacts of jet stream shifts on extreme weather events
- Improved representation in climate models

References

[Jet stream Wiki](#)

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